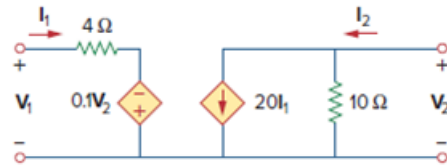


Problem

Find the y parameters for the circuit in Fig. 19.86.

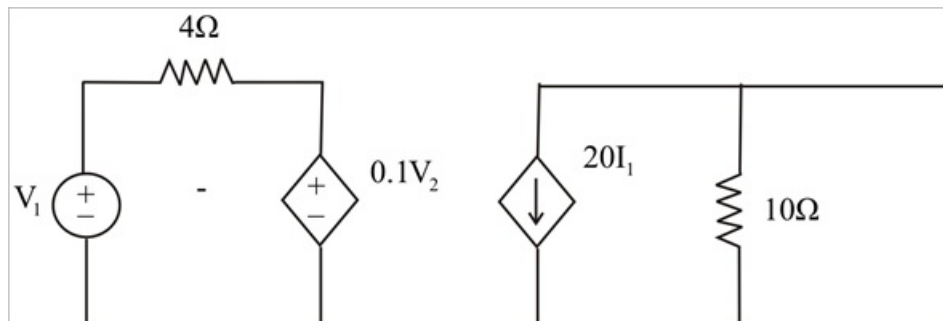
Figure 19.86



Step-by-step solution

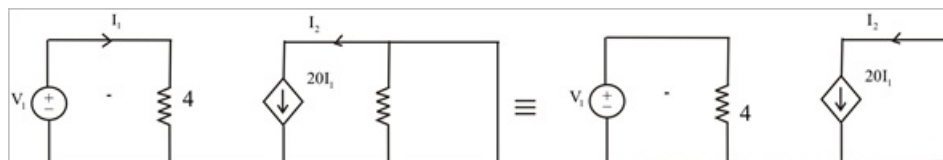
Step 1 of 5

For $V_2 = 0$ we have



Step 2 of 5

Since $V_2 = 0$ the circuit reduce to



Step 3 of 5

$$V_1 = 4I_1 \Rightarrow \frac{V_1}{I_1} = 4$$

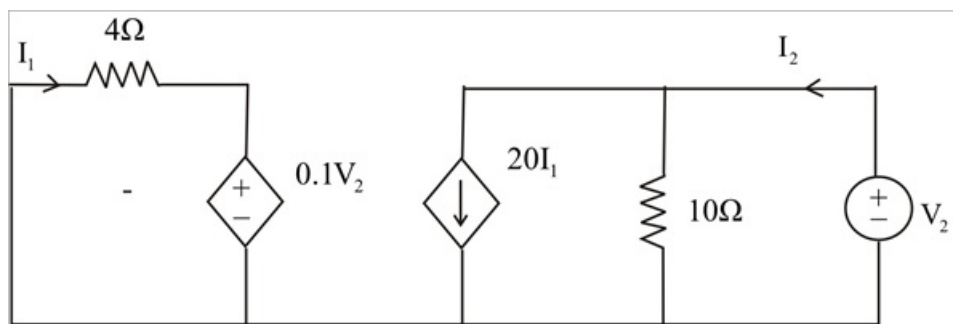
$$Y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0} = 0.25S$$

$$I_2 = 20I_1 = I_2 = 20 \times \frac{V_1}{4} \Rightarrow \frac{I_2}{V_1} = 5S.$$

$$Y_{21} = 5S.$$

For $V_1 = 0$ we have

Step 4 of 5



Step 5 of 5

From Loop I

$$I_1 = \frac{0.1V_2}{4}$$

$$\frac{I_1}{V_2} = 0.025S.$$

At Node (2)

$$I_2 = \frac{V_2}{10} + 20I_1$$

$$\Rightarrow I_2 = \frac{V_2}{10} + 20 \times \frac{0.1V_2}{4}$$

$$\frac{I_2}{V_2} = \frac{1}{10} + 0.5 = 0.6S.$$

$$Y = \begin{bmatrix} 0.25 & 0.025 \\ 5 & 0.6 \end{bmatrix} S.$$